

DRYWALL PRO



Building Code Quick Reference

Essential code requirements for drywall installation, fire ratings,
moisture protection, and sound control.

QUICK REFERENCE GUIDE

01 Stud Spacing Requirements

Proper stud spacing ensures structural integrity and prevents sagging or cracking. Always verify local codes, which may be more restrictive than the general requirements listed here.

Wall Framing

Application	Stud Spacing	Drywall Thickness	Notes
Standard load-bearing wall	16" OC	1/2"	Most common residential
Standard load-bearing wall	16" OC	5/8"	Better sound & fire performance
Non-load-bearing partition	24" OC	5/8"	Thicker board required at 24"
Non-load-bearing partition	24" OC	1/2"	Not recommended; may sag

Ceiling Framing

Application	Joist Spacing	Drywall Thickness	Notes
Standard ceiling	16" OC	1/2"	Maximum span for 1/2"
Standard ceiling	24" OC	5/8"	Required at wider spacing
Textured ceiling	16" OC	5/8"	Extra weight of texture demands thicker board
Water-based texture	16" OC	5/8"	Prevents sag from moisture absorption

KEY POINTS

- **OC** means "on center" — measured from the center of one stud to the center of the next.
- Never hang 1/2" drywall on ceilings with joists spaced more than 16" apart.
- For garage ceilings that require fire rating, use 5/8" Type X regardless of joist spacing.

02 Fastener Requirements

Correct screw selection and spacing is critical for a secure installation that passes inspection. Using the wrong screw type or spacing is one of the most common code violations.

Screw Spacing

Location	Spacing (Field)	Spacing (Edges)	Edge Distance
Ceilings	12" OC	12" OC	Min 3/8" (1/2" recommended)
Walls	16" OC	12" OC	Min 3/8" (1/2" recommended)
Fire-rated assemblies	12" OC max	12" OC max	Per UL listing

Screw Type & Length

Framing Material	Screw Type	Min Penetration	Common Lengths
Wood studs	#6 coarse thread	5/8" into framing	1-1/4" (1/2" board), 1-5/8" (5/8" board)
Steel studs (20–25 ga)	#6 fine thread	3/8" through steel	1" (1/2" board), 1-1/4" (5/8" board)
Double-layer (base)	#6 coarse or fine	5/8" into framing	1-1/4"
Double-layer (face)	#6 coarse or fine	5/8" into framing	1-5/8" to 2"

 Screws must be driven so the head is slightly below the surface without breaking the paper face. Over-driven screws that tear the paper do not hold and must be supplemented with an additional screw placed 1.5" away.

PRO TIPS

- Use a drywall screw gun with an adjustable clutch to prevent over-driving.
- Nails are generally not accepted for ceiling applications in modern codes.
- Adhesive (construction adhesive on studs) can reduce the number of fasteners needed in some assemblies, but check your local code.

03 Fire-Rated Assemblies

Fire-rated assemblies must be built exactly as tested and listed. Substituting materials, changing stud spacing, or missing firestop details can void the rating and violate code.

Common Fire-Rated Wall Assemblies

Rating	Assembly	Drywall Type	Key Requirements
1-Hour	Single layer each side	5/8" Type X	Screws 12" OC; joints staggered
2-Hour	Double layer each side	5/8" Type X	Base layer + face layer; joints offset 16" min
1-Hour (ceiling)	Single layer below joists	5/8" Type X	Screws 12" OC; no recessed lights unless IC-rated
45-Minute	Single layer each side	5/8" Type X	Wood studs 16" OC; specific UL design

Type X Drywall

- **Core composition:** Special glass-fiber reinforced gypsum core with vermiculite additives that resist heat transfer.
- **Thickness:** 5/8" is standard for Type X. 1/2" Type X exists but provides only a 45-minute rating.
- **Marking:** Must be stamped "Type X" on the back. Regular 5/8" drywall is NOT a substitute.

Penetration Firestopping

- **Electrical boxes:** Use putty pads on the back of each box in fire-rated walls.
- **Pipes & conduit:** Firestop caulk or intumescent collars required at every penetration.
- **HVAC ducts:** Fire dampers required where ducts pass through fire-rated assemblies.
- **General rule:** Any opening that breaches the fire barrier must be sealed with a listed firestop system.



UL-listed assemblies must match the tested design exactly. Do not add, remove, or substitute any component — including insulation type and density — without verifying the specific UL listing allows it.

04 Moisture Resistance

Selecting the right board type for moisture exposure prevents mold growth, structural damage, and costly callbacks. Code requirements vary by climate zone and application.

Board Selection by Application

Material	Use For	Do NOT Use For	Code Notes
Standard drywall	Dry interior walls & ceilings	Bathrooms, kitchens, basements	No moisture resistance
Green board (MR drywall)	Bathroom walls (non-wet), laundry rooms, kitchens	Shower/tub walls, steam rooms	Moisture-resistant, not waterproof
Cement board	Behind tile in showers, tub surrounds, steam rooms	N/A — suitable for all wet areas	Required by code behind tile in wet areas
Mold-resistant (purple board)	Bathrooms, basements, laundry, kitchens	Direct water immersion	ASTM D3273 score of 10 (highest)
Glass mat sheathing	Exterior sheathing, high-humidity areas	Finished interior walls (no paper face)	No paper = no mold food source

Vapor Barriers

- **Cold climates (zones 5+):** Vapor retarder required on the warm-in-winter side (interior) of exterior walls.
- **Class I:** Polyethylene sheeting (6 mil) — most restrictive, required in very cold climates.
- **Class II:** Kraft-faced insulation — acceptable in most cold climates.
- **Class III:** Latex paint — acceptable in moderate climates with adequate exterior ventilation.
- **Hot/humid climates:** Vapor retarder goes on the exterior side or is omitted; check local codes carefully.

COMMON MISTAKES

- Using green board behind shower tile — it will fail. Cement board is required.
- Installing a vapor barrier on both sides of a wall — moisture gets trapped and causes mold.
- Using standard drywall on basement walls without a moisture management plan.

05 Sound Transmission

Sound Transmission Class (STC) measures how well a wall or floor/ceiling assembly blocks airborne sound. Higher STC means better sound isolation. Building codes set minimums for multifamily and mixed-use construction.

STC Ratings by Assembly Type

Wall Assembly	STC Rating	Sound Reduction
Single layer 1/2" drywall, no insulation	33–35	Normal speech audible
Single layer 1/2" drywall + batt insulation	39–43	Loud speech partially audible
5/8" drywall + batt insulation	40–45	Loud speech faintly audible
Resilient channel + insulation	45–52	Loud speech barely audible
Double layer drywall + resilient channel	50–55	Most sounds inaudible
Staggered stud wall + insulation	50–55	Most sounds inaudible
Double stud wall + insulation	55–60+	Excellent isolation

Code Minimums (IBC/IRC)

Separation Type	Minimum STC	Minimum IIC
Walls between dwelling units	50	N/A
Floor/ceiling between dwelling units	50	50
Walls between units and public spaces	50	N/A
Floor/ceiling between units and public spaces	50	50

SOUND CONTROL TIPS

- **Resilient channel:** Decouples drywall from studs, breaking the sound vibration path. Must be installed correctly — short-circuiting with a misplaced screw voids the benefit.
- **Seal all gaps:** Even a 1% gap in a wall can reduce the STC by up to 10 points. Use acoustical sealant at all perimeters.
- **Back-to-back outlets:** Stagger electrical boxes on opposite sides of a wall by at least one stud bay — never install them in the same cavity.
- **IIC (Impact Insulation Class):** Measures footfall and impact noise through floors. Code minimum 50 for multifamily.

NEXT STEP

Put This Knowledge to Work on Your Next Project

You now have a complete quick reference for the building codes that matter most in drywall installation. Print this guide and keep it on the job site for fast lookups during framing, hanging, and inspections.

1

Print this guide for
your job site binder

2

Review your local
amendments and add
notes

3

Apply the specs on
your very next install

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